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Gated Data-Space Inversion for Reservoir Prediction

Thiago Tomás de Paula

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Previsão de Reservatórios usando Inversão do Espaço de Dados Ponderada

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Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

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Prof. Dr. Eugênio Libório Feitosa Fortaleza
UnB/FT/ENM
Orientador

Examinador interno

Examinador interno

Abstract

Accurate prediction of reservoir performance under uncertainty is essential for decision-making in geological reservoir management. Data-Space Inversion (DSI) (Sun; Durlofsky, 2017) provides an attractive alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations. Recently, Lima, Emerick, and Ortiz (2020) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. Despite these advantages, DSI-ESMDA, like other ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty. Methods such as covariance localization help to ameliorate this issue, but variance underestimation is still observed in practical applications. One contributing factor to this progressive loss of variance is that small production mismatches, often dominated by measurement noise or sampling effects, may trigger repeated micro-updates that reduce ensemble diversity without improving predictive skill. This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems (Fortaleza *et al.*, 2022) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing (Donoho; Johnstone, 1994; Donoho, 1995). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The approach is evaluated on multiple synthetic reservoirs and one real field application using production rates from producers and injectors. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

Keywords: ; ; .

Resumo

Palavras-chave: ; ; .

Contents

1 Introduction

1.1 Work overview

The economic success of oil and gas fields is increasingly contingent on the quality of reservoir management decisions, which are made under uncertainty and based on limited data. Kalman-based ensemble methods are widely used to quantify uncertainty in reservoir forecasts by adjusting reservoir models to match production data, but require vast computational resources to perform repeated reservoir simulations. In an effort to reduce the computational burden, Data-Space Inversion (DSI) (Sun; Durlofsky, 2017) was introduced as an alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations.

Lima, Emerick, and Ortiz (2020) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. However, DSI-ESMDA, like related ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty.

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. Findings on both synthetic and real reservoirs show that Gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match.

Gated DSI-ESMDA acts as a stabilizing extension rather than a replacement, and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The power of this contribution is that it provides a simple, yet effective means of improving the robustness of ensemble-based inversion methods without requiring additional simulations or complex modifications to existing workflows.

1.2 Background context

The economic viability of oil and gas operations depends critically on sound reservoir management, where decisions must be made under significant uncertainty and with inherently limited subsurface data. Accurate characterization of this uncertainty is essential: overconfident forecasts can lead to poor investment decisions, while excessively uncertain ones impede operational planning. History matching — the process of calibrating reservoir models to observed production data — has long been central to reducing forecast uncertainty, and ensemble-based methods have become the dominant paradigm for performing this calibration at scale.

Kalman-based ensemble methods, such as the Ensemble Smoother with Multiple Data Assimilation (ES-MDA), address this problem by maintaining an ensemble of plausible reservoir models and updating them iteratively as production data are assimilated. By propagating uncertainty through an ensemble rather than relying on a single deterministic model, these methods provide a principled framework for quantifying forecast uncertainty. However, they carry a substantial computational cost: each update cycle requires running full reservoir simulations for every ensemble member, making large-scale uncertainty quantification expensive and, in practice, often infeasible.

Data-Space Inversion (DSI), introduced by Sun and Durlofsky (2017), was proposed as a computationally efficient alternative to traditional model-space history matching. Rather than updating subsurface model parameters and re-running simulations, DSI operates directly on the ensemble of simulated production forecasts — working entirely in data space. This eliminates the need for repeated reservoir simulations during the assimilation process, dramatically reducing the computational burden while still yielding calibrated posterior forecast distributions.

Building on this foundation, Lima, Emerick, and Ortiz (2020) developed DSI-ESMDA, which replaces the Randomized Maximum Likelihood minimizations of the original DSI formulation with an iterative ES-MDA-style Kalman update. This modification brings further gains in computational efficiency and produces competitive posterior forecasts, establishing DSI-ESMDA as a practical and attractive framework for reservoir uncertainty quantification.

Despite these advances, DSI-ESMDA inherits a well-known vulnerability of ensemble smoother methods: excessive variance loss during data assimilation. When observations strongly constrain the ensemble, the spread of forecasts can collapse beyond what is physically justified, leading to an underestimation of posterior uncertainty. This phenomenon is particularly problematic in reservoir forecasting, where robust uncertainty envelopes are essential for risk-aware decision-making. The issue is compounded in data-space methods, where the ensemble is not replenished through new simulations and the initial ensemble spread is the only source of diversity throughout the assimilation process.

Mitigating variance collapse without sacrificing data match quality or introducing additional computational overhead remains an open challenge. Existing remedies — such as inflation schemes and localization — typically require careful tuning and may not transfer cleanly to the data-space setting. This motivates the development of a principled, lightweight mechanism that can stabilize the Kalman update in DSI-ESMDA, preserving ensemble diversity and producing more reliable uncertainty quantification across a range of reservoir scenarios.

1.3 Research problem

Literature on DSI and DSI-ESMDA take different approaches to mitigating variance loss, but they are either computationally expensive or require careful tuning of hyperparameters. Ideally, a solution would be simple to implement, require minimal tuning, and be computationally efficient. This work proposes introducing a straightforward gating mechanism on DSI-ESMDA's innovation vector, which selectively suppresses the influence of well-matched observations while preserving the impact of significant mismatches. Gated DSI-ESMDA thus modulates the posterior ensemble variance with a single parameter, the intensity of the gating function.

1.4 Research questions

This work addresses the following research questions:

1. Can a gating mechanism be introduced into DSI-ESMDA to selectively modulate the influence of observations based on their match quality?
2. Does the proposed innovation gating reduce variance loss during assimilation and produce more robust uncertainty envelopes in the prediction phase without degrading the data match?
3. How does the performance of Gated DSI-ESMDA compare to standard DSI-ESMDA across synthetic and real reservoir case studies?

1.5 Significance

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems (Fortaleza *et al.*, 2022) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing (Donoho; Johnstone, 1994; Donoho, 1995). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The proposed method inherits the computational efficiency of DSI-ESMDA while providing a robust, easily tunable mechanism for modulating posterior distribution variance, setting it as a practical and effective tool for uncertainty quantification in reservoir forecasting.

1.6 Methodology overview

The approach is evaluated against the original DSI-ESMDA on two synthetic reservoirs and one real field application using production rates from producers and injectors. The synthetic reservoirs are the Egg and Olympus ensembles, which are widely used in the literature to benchmark history matching and uncertainty quantification methods, while the real field application is a mature oil field in the Brazilian pre-salt. Posterior and prior ensembles are compared across several metrics, including observation coverage and mean squared error, to assess the impact of innovation gating on variance loss and uncertainty quantification. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

1.7 Dissertation outline

Chapter 2 presents a review of the literature on reservoir uncertainty quantification, focusing on ensemble-based methods and data-space inversion. Chapter 3 introduces the theoretical framework underpinning the proposed methodology, from the relevant elements of reservoir engineering as a whole to the mathematical formulation of DSI-ESMDA and the innovation gating mechanism. Chapter 4 details the methods used to evaluate the proposed approach, including descriptions of the synthetic and real reservoir case studies, data collection, and analysis techniques. Chapter 5 presents the results of applying innovation-gated DSI-ESMDA to the case studies, including quantitative and qualitative assessments of performance relative to standard DSI-ESMDA. Chapter 6 discusses the implications of the findings and their relationship to the existing literature. Chapter 7 summarizes the main contributions and provides final remarks on the research and future work.

2 Literature Review

2.1 Chapter outline

Aims and structure

The DSI method introduced by Sun and Durlofsky [419] employs PCA to reparameterize the predicted data from the prior ensemble into a lower- dimensional space and uses RML to sample from the posterior distribution of predicted data. The authors applied a data transformation prior to PCA to enhance linearity. Sun et al. [421] further extended the DSI method by introducing a more generalized data transformation procedure, and Sun and Durlofsky [420] applied the method to a geological carbon storage problem. Earlier, Scheidt et al. [381] proposed a similar concept in a method called prediction-focused analysis (PFA), which involves projecting prior predic- tions into a low-dimensional space and using kernel smoothing to estimate the joint distribution of historical and forecasted data. However, PFA seems to be limited to problems with a small number of data points, as it re- quires projection into very few dimensions (two or three in the examples presented). Satija and Caers [379] improved the PFA method by incorpo- rating canonical functional component analysis to enhance linearity in the projected data, and Satija and Caers [380] applied this approach to a reser- voir problem, finding that it provided uncertainty estimates for production forecasts in reasonable agreement with rejection sampling. Lima et al. [263] combined ES-MDA and DSI, resulting in a more robust and efficient DSI implementation. This method has been further enhanced through the use of various deep learning techniques, leading to successful applications in naturally fractured reservoirs [218] and geological carbon storage [469, 217] (Emerick, 2025, section D.8).

2.2 Chapter summary

Key takeaways

3 Theory framework

3.1 Chapter outline

Aims and structure

3.2 Elements of reservoir engineering

3.2.1 Reservoir formation

3.2.2 Reservoir simulation

3.3 Kalman Filtering

3.4 Data-Space Inversion

Let $\mathbf{d}$ denote the true production data vector of a reference reservoir across time, and let $\mathbf{d}_j$ denote the production data vector produced by simulating a realization of the reference reservoir using model parameters $\mathbf{m}_j$. For an ensemble of N_e realizations, we have N_e different such vectors, each of which dependent on its own parameters vector. All data vectors are assumed whole, i.e., without missing entries, as well as sinchronized.

Of the original reference data vector, only part of it is used in the history matching process. Denote this vector by $\mathbf{d}_h$. Following the same pattern, $\mathbf{d}_{h,\,j}$ denotes the history match period data output by member j of the ensemble. Assume $\mathbf{d}_h$ is corrupted by additive measurement noise $\mathbf{e}_d \sim N(\mathbf{0}, \mathbf{C}_e)$, where $\mathbf{C}_e$ is the noise covariance matrix. The resulting vector is the oberved data during the history period, denoted $\mathbf{d}_{\text{obs}}$. DSI-ESMDA updates the ensemble outputs by iterating a Kalman-like data assimilation, shown below (Lima; Emerick; Ortiz, 2020):

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + (\mathbf{R} \circ \mathbf{K}^k)(\mathbf{d}_{\text{obs}} + \sqrt{\alpha_k}\mathbf{e}_j^k - \mathbf{d}_{h,\,j}^k), \tag{3.1}$$

with $1 \le k \le N_a$, N_a the number of assimilations DSI-ESMDA must do. Thus, $\mathbf{d}_j^k$ for $k = 1$ is the ensemble prior $\mathbf{d}_j$, and $\mathbf{d}_j^{Na+1}$, the posterior. Vector $\mathbf{e}_j^k \sim N(\mathbf{0}, \mathbf{C}_e)$ is a random disturbance, and α_k coefficients are positive constants that satisfy $\sum_k 1/\alpha_k = 1$.

Matrices $\mathbf{R}$ and $\mathbf{K}$ are the localization and the modified Kalman gain, respectively. $\mathbf{K}$ is defined as

$$\mathbf{K}^k = \Delta\mathbf{D}^k(\Delta\mathbf{D}_h^k)^\top \left(\Delta\mathbf{D}_h^k(\Delta\mathbf{D}_h^k)^\top + \alpha_k\mathbf{C}_e\right)^{-1}, \tag{3.2}$$

with

$$\Delta \mathbf{D} = \frac{1}{\sqrt{N_e - 1}} \begin{bmatrix} \mathbf{d}_1 - \overline{\mathbf{d}} & \cdots & \mathbf{d}_{N_e} - \overline{\mathbf{d}} \end{bmatrix}, \tag{3.3}$$

$$\overline{\mathbf{d}} = \frac{1}{N_e} \sum_{j=1}^{N_e} \mathbf{d}_j, \tag{3.4}$$

a matrix that quantifies the spread of the ensemble. $\Delta \mathbf{D}_h$ is defined analogously by constraining the above equations to the history matching period:

$$\Delta \mathbf{D}_h = \frac{1}{\sqrt{N_e - 1}} \begin{bmatrix} \mathbf{d}_{h,1} - \overline{\mathbf{d}}_h & \cdots & \mathbf{d}_{h,N_e} - \overline{\mathbf{d}}_h \end{bmatrix}, \tag{3.5}$$

$$\overline{\mathbf{d}}_h = \frac{1}{N_e} \sum_{j=1}^{N_e} \mathbf{d}_{h,j}. \tag{3.6}$$

R adjusts the Kalman gain, preserving correlated measures while diminishing the impact of spuriously correlated measures. As such, its entries lie in $[0,1]$, and pointwisely multiply **K**. Intuitively, a correlation is strong if measures of a same quantity were taken closer in space and/or in time, and weaker otherwise. Denoting this "distance" by δ_{uv}, where u is an index from $\mathbf{d}_j$ and v an index from $\mathbf{d}_h$, **R** is given by

$$R_{uv} = r(\delta_{uv}), \tag{3.7}$$

$$r(\delta) = \begin{cases} -\frac{1}{4}\delta^5 + \frac{1}{2}\delta^4 + \frac{5}{8}\delta^3 - \frac{5}{3}\delta^2 + 1, & 0 \le \delta \le 1 \\ \frac{1}{12}\delta^5 - \frac{1}{2}\delta^4 + \frac{5}{8}\delta^3 + \frac{5}{3}\delta^2 - 5\delta + 4 - \frac{2}{3}\delta^{-1}, & 1 < \delta \le 2 \\ 0, & \delta > 2 \end{cases}, \tag{3.8}$$

$$\delta_{uv} = \sqrt{\left(\frac{\Delta x'_{uv}}{L_{x'}}\right)^2 + \left(\frac{\Delta y'_{uv}}{L_{y'}}\right)^2 + \left(\frac{\Delta t_{uv}}{T_c}\right)^2}, \tag{3.9}$$

$$\begin{bmatrix} \Delta x'_{uv} \\ \Delta y'_{uv} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \Delta x_{uv} \\ \Delta y_{uv} \end{bmatrix} \tag{3.10}$$

Δx_{uv} and Δy_{uv} are the x and y differences, respectively, between the points in space from where measure u and v were taken. In this context, measurements always come from a well, and so these differences are always distances between wells. Δt_{uv} is the difference between the time when measurement u was taken, and the time when v was taken. Note that while v is restrained to the history matching period by definition, u is not. The inclusion of Δt_{uv} guarantees that measurements far apart in time will hold little correlation, even if taken from the same well.

A rotation by θ is applied to take into account the anisotropy of reservoir permeability, and are $L_{x'}$, $L_{y'}$, T_c positive constants that regulate the critical points of differences in x, y and t separately. By critical, it is understood the inflection point $\delta = 1$ in $r(\delta)$. The greater

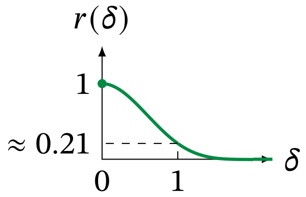


Figure 3.1 – Compact correlation function r used to build the localization matrix $\mathbf{R}$.

Figure 3.2 – Illustration of the localization applied in several different measurements. *Left panel:* Localization between measurements from the same well but from different times. *Center panel:* Localization between measurements from the different wells but at the same times. *Right panel:* Localization between measurements from the different wells at different times.

the critical value, the more lax the localization, allowing more weakly correlated changes to hold more impact.

Localization greatly improves the quality of the posterior, but if the prior zeroes in on an average that is too different from the reference, it is hard to adjust it in order to inject some variance on the collapsed curves. This is partly the reason behind the random disturbance in: injecting variance on the ensemble fit.

Even when the localization $\mathbf{R}$ is well tuned (i.e., θ, $L_{x'}$, $L_{y'}$ and T_c generate adequate posteriors), DSI-ESMDA may suffer the inherent history matching problem of making poor predictions despite good fit to past data. The ensemble posterior narrows around a curve that is not representative of the reservoir.

The inherent problem of history matching may be impossible to solve, but it may be mitigated by allowing the posterior to spread more. We propose doing so by adaptively adjusting the importance of the history mismatch when running the update iterations:

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + (\mathbf{R} \circ \mathbf{K}^k)[\boldsymbol{\alpha}_j^k \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,\,j}^k)], \tag{3.11}$$

$$\alpha_{j,\,v}^k = \operatorname{erf}\left(\frac{1}{\lambda\sqrt{2\sigma_v^2}}|(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,\,j}^k)_v|\right) \tag{3.12}$$

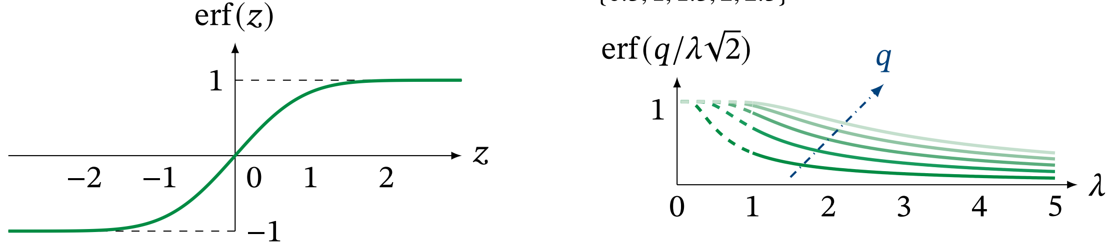
The equation above is based on the work of (Fortaleza *et al.*, 2022), and a detailed derivation is provided in appendix A. The smaller the mismatch of a measurement relative to the noise at that measure, the smaller the update. Constant $\lambda \geq 1$ inflates the deviation, further softening the Kalman-like update. At the extreme ($\lambda \to \infty$ or $\mathbf{d}_{\text{obs}} = \mathbf{d}_{h,\,j}^k$), $\boldsymbol{\alpha}_j^k = \mathbf{0}$ and the DSI update is fully undone.

3.5 Chapter summary

Figure 3.3 – Important functions for understanding the proposed Data-Space Inversion algorithm.

(a) Special error function $\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2}\, dt$

(b) Values of $\alpha^k_{j,\,v}$ when $|(\mathbf{d}_{\text{obs}} - \mathbf{d}^k_{h,\,j})_v| = q\sigma_v$ for $q \in \{0.5, 1, 1.5, 2, 2.5\}$

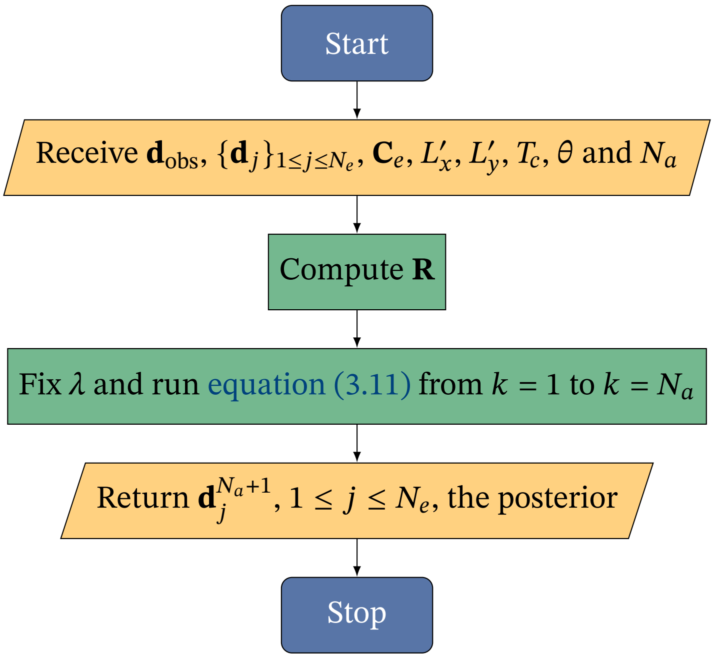


Source: Self-elaboration.

Figure 3.4 – Illustration of the effect of λ on the posterior ensemble for fixed $\mathbf{R}$ and N_a. Yellow lines are the values of $\alpha_j^{N_a+1}$ across time. *Left panel*: DSI-ESMDA prior and posterior ensembles. *Center panel*: DSI-ESMDA with $\lambda = 1$ filtering. *Right panel*: DSI-ESMDA with $\lambda = 5$ filtering. Posterior gains variance as λ grows.

Source: Self-elaboration.

Figure 3.5 – Outline of the proposed Data-Space Inversion algorithm.



Source: Self-elaboration.

Key takeaways

4 Methods

4.1 Chapter outline

Aims and structure

4.2 Reservoir case studies

Three reservoirs were used as case studies for this work, two of which are synthetic reservoirs and one is a real reservoir. This section presents them, highlighting their most significant structural and geological characteristics within the scope of this work. However, as information regarding the real reservoir is sensitive, it is only briefly discussed.

The choice of these reservoirs was based on the availability of data and their suitability for testing the proposed methods: the Egg and Olympus ensembles are widely used in the literature for benchmarking purposes (with the Olympus ensemble being the more complex of the two), while the real field provides a practical application scenario for the methods developed in this work.

4.2.1 Egg ensemble

Originally introduced as a deterministic model in a study aimed at exploring the reasons behind reservoir flooding problems and identifying conditions for optimal solutions (Zandvliet *et al.*, 2007), a stochastic version was later proposed as a benchmark in a robust waterflooding optimization problem (Van Essen *et al.*, 2009). This reservoir consists of 18,533 active cells with dimensions of 8x8x4 m. The arrangement of these cells forms an egg-shaped grid, hence the name (Jansen *et al.*, 2014).

4.2.1.1 Realizations

The standard Egg ensemble comprises a single deterministic realization and 100 permeability realizations of a channelized reservoir. In addition to eight injectors labeled INJECT1 through INJECT8, each realization includes four vertical producer wells labeled PROD1 through PROD4, which produce oil and water but not gas. Table 4.1 shows some of the properties shared by the realizations.

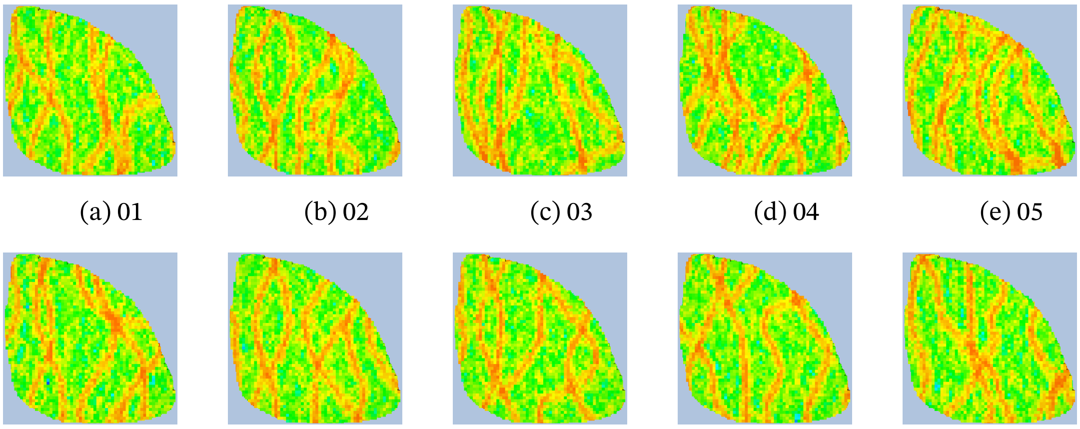
In figure 4.1, renderings of permeability along the x-axis in some of Egg's realizations are depicted, with colors assigned according to a logarithmic scale. To enhance visualization, the z-axis was magnified fivefold.

Table 4.1 – Some geometry and fluid properties of the Egg ensemble.

Property	Value	Unit
Rock Compressibility	0.0	1/Pa
Water Compressibility	1.0×10^{-10}	1/Pa
Oil Dynamic Viscosity	5.0×10^{-3}	Pa·s
Water Dynamic Viscosity	1.0×10^{-3}	Pa·s
Well Diameter	20	cm
Simulation Time	3600	days

Adapted from Jansen *et al.* (2014).

Figure 4.1 – Top view of selected realizations 1-10 from the Egg ensemble. Renderings were produced using the open-source software ResInsight.



(a) 01 (b) 02 (c) 03 (d) 04 (e) 05

(f) 06 (g) 07 (h) 08 (i) 09 (j) 10

Source: Filho *et al.* (2025).

4.2.2 Olympus ensemble

Launched in early 2017, the Olympus reservoir model was designed as a benchmark environment to problems concerning control optimization and reservoir management within the framework of the Integrated Systems Approach for Petroleum Production 2 research consortium (Fonseca *et al.*, 2018; Fonseca *et al.*, 2020). In what follows, the depiction of Olympus mirrors the one outlined by Fonseca *et al.* (2020), with adjustments made where deemed necessary.

Inspired by an oil field in the North Sea, this synthetic reservoir is approximately rectangular, measuring 9x3 km, with a depth of 50 m divided into 16 layers. The 3D grid comprises cells with dimensions of approximately 50x50x3 m, totaling 341,728 cells, of which 192,750 are active. The perimeter of the reservoir is defined by a fault, with six additional smaller faults present within it. Furthermore, Olympus features two distinct zones separated by an impermeable shale layer: The upper zone contains fluvial channel sands embedded in floodplain shales, while the lower zone consists of alternating layers of coarse, medium, and fine sands that are inclined relative to the general structural dip of the field. The reservoir includes four facies types in different layers (see table 4.2), and geological properties, such as porosity, permeability, and the net-to-gross ratio, were generated using standard geostatistical techniques for the different facies types. In the absence of sufficient early-stage data, no porosity/permeability ratio was used. Permeabilities in the x and y directions were identical, while in the z direction, it corresponded to 10% of that in the x direction.

Table 4.2 – Types and distribution range intervals of facies properties in Olympus.

Type	Presence zones	NG	Porosity	Permeability (μm^2)
Channel sand	Upper	0.8-1	0.2-0.35	0.4-1
Shale	Upper, perimeter	0.0	0.03	0.001
Coarse sand	Lower	0.7-0.9	0.2-0.3	0.15-0.4
Medium sand	Lower	0.75-0.95	0.1-0.2	0.075-0.15
Fine sand	Lower	0.9-1	0.05-0.1	0.01-0.05

Source: Fonseca *et al.* (2020).

Based on available exploration well records, it was determined that the oil/water contact depth was 2092 m, with an in-situ hydrostatic pressure of 205 bar. The initial water saturation distribution was modeled using capillary pressure curves. Different capillary pressure curves were assigned to different facies types. Therefore, changes in the facies resulted in different initial water saturation distributions. The reservoir's fluid was similar to oil without dissolved gas, and some of its properties are summarized in table 4.3.

Table 4.3 – Olympus' fluid properties.

Property	Value	Unit
Rock compressibility	1.42×10^{-10}	1/Pa
Water compressibility	3.97×10^{-10}	1/Pa
Oil dynamic viscosity	$2.8\text{-}3.5 \times 10^{-3}$	Pa·s
Water dynamic viscosity	0.398×10^{-3}	Pa·s
Well diameter	19	cm
Simulation Time	7380	days

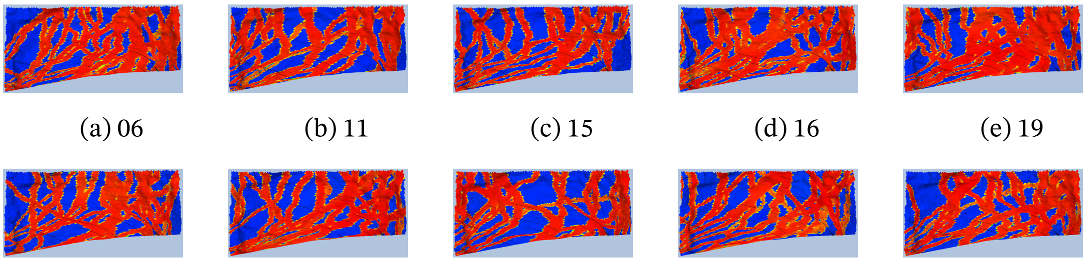
Source: Adapted from Fonseca *et al.* (2020).

4.2.2.1 Realizations

A set of 50 realizations was generated by varying facies through random seed changes, while keeping consistent mesh geometry, faults, and oil-water contact. Uncertain properties such as facies, porosity, permeability, NG ratio, initial water saturation distribution, and transmissibility within faults were considered.

By first generating a high-fidelity base model with approximately 5 million grid cells, 5 wells were drilled into this base case model and synthetic records were generated for each of these wells. These records were then used to constrain the generation of 50 high-fidelity realizations to capture uncertainty. Each was upscaled for flow simulations using the flow-based upscaling method, featuring a total of 18 wells in each realization, of which 11 are producers and 7 are injectors. Each realizations includes 7 injector wells, labeled INJ-1 to INJ-7, and 11 producer wells, labeled PROD-1 to PROD-11. Producers produce oil and water, but not gas. Figure 4.2 illustrates the permeability in the x direction for selected realizations, depicted with colors on a logarithmic scale and a fivefold exaggeration of the z-axis.

Figure 4.2 – Top view of selected realizations from the OLYMPUS ensemble. Renderings were produced using the open-source software ResInsight.



(a) 06 (b) 11 (c) 15 (d) 16 (e) 19

(f) 25 (g) 32 (h) 38 (i) 43 (j) 44

Source: Filho *et al.* (2025).

4.2.3 Real field

Given that most parameters characterizing the properties of the carbonate reservoir in question cannot be disclosed due to ongoing confidentiality agreements with industrial partners, only essential information is provided.

The real field is a mature reservoir located in the Brazilian pre-salt, with a production history spanning over 25 years. It is comprised of 8 injector wells, labeled I1 to I8, and 16 producer wells, labeled P1 to P16. Producers produce oil and water, but not gas.

An ensemble of 200 realizations was generated from observed production data, with a total of 510 simulation time steps of roughly 60 days each. Importantly, observed data has a time vector that is not identical to the simulation's but is a subset of it.

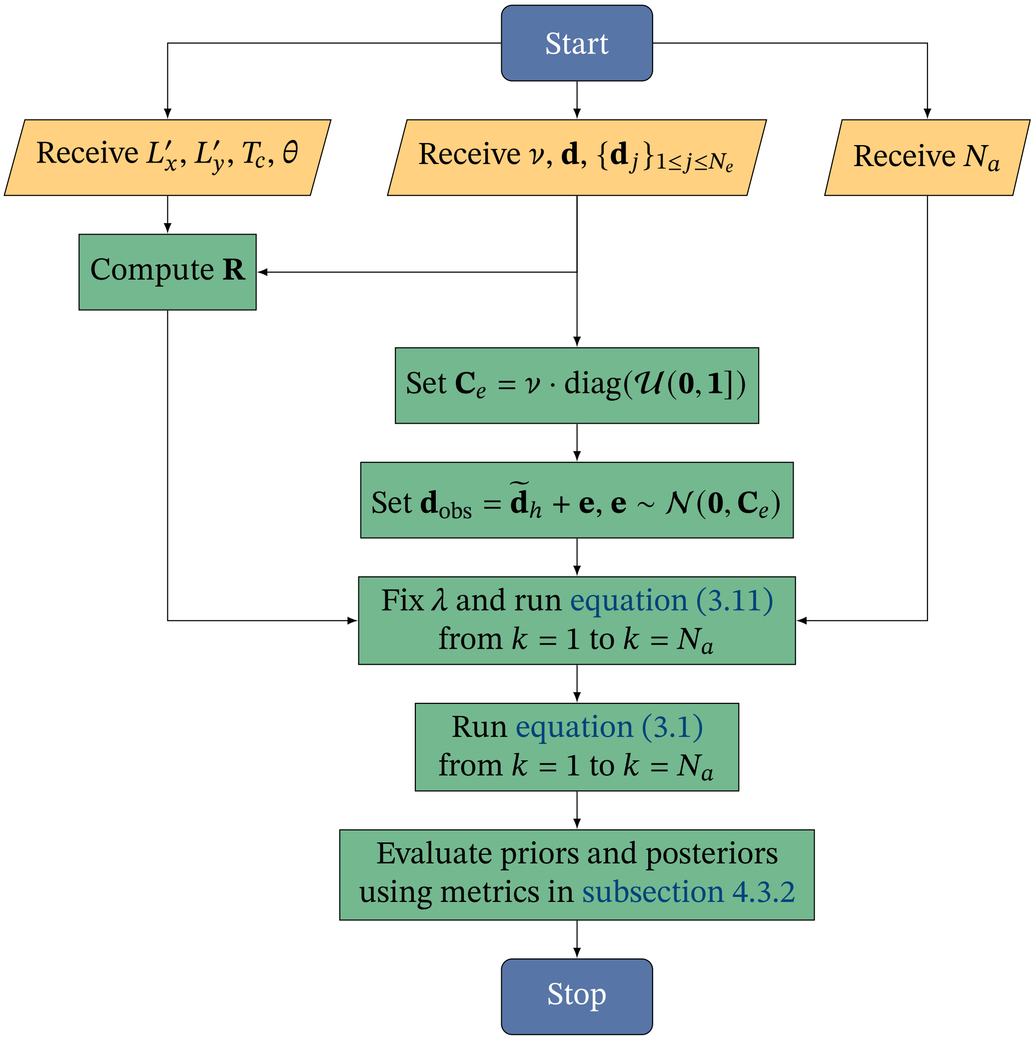
4.3 Evaluation workflow and parameters

4.3.1 Evaluation workflow

While the outline of both DSI-ESMDA and the proposed Gated DSI-ESMDA methods have been presented in chapter 3, it remains to be explained how the error covariance matrix $\mathbf{C}_e$ is constructed, what parameters are used, and what metrics will be used to evaluate the performance of the methods.

Figure 4.3 illustrates the workflow of the Evaluation process by detailing figure 3.5 in chapter 3.

Figure 4.3 – Outline of the evaluation process.



Source: Self-elaboration.

First, all data is normalized to the range $[0,1]$ using the minimum and maximum values of the groundtruth production data, as follows:

$$\widetilde{\mathbf{d}}_{j,f} = \frac{\mathbf{d}_{j,f} - \min(\mathbf{d}_f)}{\max(\mathbf{d}_f) - \min(\mathbf{d}_f)} \tag{4.1}$$

$$\widetilde{\mathbf{d}}_{f} = \frac{\mathbf{d}_{f} - \min(\mathbf{d}_f)}{\max(\mathbf{d}_f) - \min(\mathbf{d}_f)}, \tag{4.2}$$

where index j denotes the realization, f denotes the production data feature, and the tilde indicates normalization. This process is repeated for all features and realizations to improve the conditioning of the data and facilitate noise modeling, as shown next. An important caveat is that the normalization in equations (4.1) and (4.2) will not work if the groundtruth feature is constant, so such features are removed if detected. The error covariance matrix $\mathbf{C}_e$

is assumed to be diagonal, with its non-zero entries uniformly sampled from $(0, 1]$ and scaled by a constant $\nu > 0$. This constant is named the noise magnitude and is a hyperparameter of the Evaluation. However, ν is only means to easier explore the effects of different noise magnitudes on the performance of the methods, and not an essential parameter of the methods themselves.

Next, observed data is constructed by adding noise to the normalized groundtruth data restricted to the history period:

$$\widetilde{\mathbf{d}}_{\text{obs}} = \widetilde{\mathbf{d}}_h + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_e). \tag{4.3}$$

Since real field data already contains noise, this step is skipped for that case, and observed data is simply the normalized groundtruth data restricted to the history period.

Finally, the methods are applied to the pre-processed data, and their priors and posteriors are evaluated using the metrics described in subsection 4.3.2.

4.3.2 Evaluation metrics

Both the prior and posterior ensembles are evaluated on the history and forecast periods using the following metrics (Emerick, 2025, sections 9.4.2 and 9.4.4):

- observation coverage (OC),
- mean squared error (MSE),
- data-mismatch objective function (denoted O).

OC measures the percentage of observed data that falls within the range of an ensemble, while MSE quantifies the average squared difference between the observed and predicted data. DMOF is a metric that quantifies the discrepancy between observed and predicted data. Mathematically, each metric is defined as follows:

$$\text{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1}\left(\min(\mathbf{Y})_i \leq d_{h,i} \leq \max(\mathbf{Y})_i\right), \tag{4.4}$$

$$\text{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1}\left(\min(\mathbf{Y})_i \leq d_{v,i} \leq \max(\mathbf{Y})_i\right), \tag{4.5}$$

$$\text{MSE}(\mathbf{Y}) = \frac{1}{N_h}(\mathbf{d}_h - \overline{\mathbf{y}})^\top \mathbf{C}^{-1}(\mathbf{d}_h - \overline{\mathbf{y}}), \quad \mathbf{C} = \Delta\mathbf{Y}\Delta\mathbf{Y}^\top + \mathbf{C}_e \tag{4.6}$$

$$O(\mathbf{y}_j) = \frac{1}{2N_h}(\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1}(\mathbf{d}_h - \mathbf{y}_{j,h}), \tag{4.7}$$

where

- $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_{N_e}]$ denotes an ensemble (posterior or prior),
- N_h is the number of data points in the history period,
- N_v is the number of data points in the validation period,

- $\bar{\mathbf{y}}$ is the mean of the ensemble,
- $\max(\mathbf{Y})_i$ is the maximum value in the ensemble at datum i,
- $\min(\mathbf{Y})_i$ is the minimum value in the ensemble at datum i,
- $\Delta\mathbf{Y}$ is the deviation of the predicted data from its mean, see equation (3.3).
- $\mathbb{1} : \{0,1\} \to \{0,1\}$ returns 1 if the condition in its argument is true and 0 otherwise.

Both history and validation splits of groundtruth $\mathbf{d}$ are used to compute observation coverages, while MSE and misfit exist only for the history period. Despite the similarity between the MSE and the misfit, the latter is computed for each realization in the ensemble, while the former is computed for the entire ensemble. In this work, the misfit will be presented as an average over all realizations in the ensemble, with an associated standard deviation.

4.3.3 Evaluation parameters

Table 4.4 – Time-related simulation parameters per reservoir.

Reservoir	Number of realizations	Number of time steps	Time step size	Simulation life	History split
Egg	100	120	30 days	10 years	60%
Olympus	49	20	1 year	20 years	60%
Real field	200	510	60 days	30 years	60%

Source: Self-elaboration.

Table 4.5 – Localization-related simulation parameters per reservoir.

Reservoir	Critical length in x	Critical length in y	Critical time	Anisotropy angle
Egg	100	100	3000 days	0°
Olympus	100	100	7200 days	0°
Real field	-	-	10500 days	0°

Source: Self-elaboration.

Table 4.6 – Evaluation hyperparameters.

Parameter	Symbol	Value	Unit
Kalman-like iterations	N_a	1	-
Noise magnitudes	ν	$\{0.001, 0.01\}$	-
Gating intensities	λ	$\{1, 2, 4, 8, 16\}$	-

Source: Self-elaboration.

4.4 Chapter summary

Key takeaways

5 Results

5.1 Chapter outline

Aims and structure

5.2 Egg ensemble

5.3 Olympus ensemble

5.4 Real field

5.5 Chapter summary

Key takeaways

Table 5.1 – Summary of the Egg ensemble DSI-ESMDA results.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation				Avg.		Std. Dev.	
		Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.
10	No	0.92	0.34	1.00	1.00	1.03	0.96	9.63	0.50	5.05	0.01
	Yes	0.92	0.48	1.00	1.00	1.03	0.96	9.63	0.55	5.05	0.03
20	No	0.93	0.36	1.00	0.98	0.97	0.93	6.93	0.49	3.32	0.00
	Yes	0.93	0.52	1.00	1.00	0.97	0.93	6.93	0.53	3.32	0.02
30	No	0.87	0.36	0.93	1.00	0.99	0.94	16.77	0.49	7.01	0.00
	Yes	0.87	0.53	0.93	0.90	0.99	0.94	16.77	0.57	7.01	0.03
40	No	0.94	0.35	1.00	1.00	1.01	0.96	8.62	0.50	4.79	0.00
	Yes	0.94	0.52	1.00	1.00	1.01	0.96	8.62	0.55	4.79	0.02
50	No	0.92	0.35	1.00	0.90	1.00	0.94	10.70	0.50	5.29	0.01
	Yes	0.92	0.53	1.00	1.00	1.00	0.94	10.70	0.55	5.29	0.03
60	No	0.92	0.36	0.99	0.94	0.99	0.94	9.13	0.49	5.14	0.00
	Yes	0.92	0.53	0.99	0.98	0.99	0.94	9.13	0.55	5.14	0.03
70	No	0.91	0.34	1.00	1.00	1.03	0.95	8.62	0.50	4.08	0.01
	Yes	0.91	0.53	1.00	1.00	1.03	0.95	8.62	0.55	4.08	0.02
80	No	0.91	0.36	1.00	0.90	1.01	0.94	26.01	0.50	19.20	0.01
	Yes	0.91	0.56	1.00	1.00	1.01	0.94	26.01	0.58	19.20	0.04
90	No	0.87	0.36	0.98	0.96	1.01	0.95	43.29	0.50	31.62	0.00
	Yes	0.87	0.55	0.98	1.00	1.01	0.95	43.29	0.59	31.62	0.05
100	No	0.93	0.35	1.00	1.00	0.98	0.93	7.25	0.49	3.45	0.00
	Yes	0.93	0.52	1.00	1.00	0.98	0.93	7.25	0.53	3.45	0.02

Source: Self-elaboration.

Table 5.2 – Summary of the Egg ensemble Gated DSI-ESMDA results.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation				Avg.		Std. Dev.	
		Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.
10	No	0.92	0.41	1.00	1.00	1.03	0.97	9.63	0.52	5.05	0.01
	Yes	0.92	0.57	1.00	1.00	1.03	0.97	9.63	0.60	5.05	0.02
20	No	0.93	0.43	1.00	1.00	0.97	0.94	6.93	0.51	3.32	0.01
	Yes	0.93	0.58	1.00	1.00	0.97	0.94	6.93	0.58	3.32	0.02
30	No	0.87	0.39	0.93	0.99	0.99	0.95	16.77	0.51	7.01	0.01
	Yes	0.87	0.55	0.93	0.90	0.99	0.95	16.77	0.61	7.01	0.03
40	No	0.94	0.42	1.00	1.00	1.01	0.97	8.62	0.53	4.79	0.01
	Yes	0.94	0.57	1.00	1.00	1.01	0.97	8.62	0.60	4.79	0.02
50	No	0.92	0.43	1.00	0.90	1.00	0.95	10.70	0.51	5.29	0.01
	Yes	0.92	0.59	1.00	1.00	1.00	0.95	10.70	0.60	5.29	0.02
60	No	0.92	0.41	0.99	0.95	0.99	0.94	9.13	0.51	5.14	0.01
	Yes	0.92	0.57	0.99	0.96	0.99	0.95	9.13	0.60	5.14	0.02
70	No	0.91	0.40	1.00	1.00	1.03	0.96	8.62	0.52	4.08	0.01
	Yes	0.91	0.57	1.00	1.00	1.03	0.97	8.62	0.60	4.08	0.02
80	No	0.91	0.40	1.00	0.93	1.01	0.95	26.01	0.50	19.20	0.01
	Yes	0.91	0.59	1.00	1.00	1.01	0.95	26.01	0.60	19.20	0.03
90	No	0.87	0.38	0.98	0.94	1.01	0.96	43.29	0.51	31.62	0.01
	Yes	0.87	0.56	0.98	0.99	1.01	0.97	43.29	0.61	31.62	0.04
100	No	0.93	0.43	1.00	1.00	0.98	0.94	7.25	0.51	3.45	0.01
	Yes	0.93	0.58	1.00	1.00	0.98	0.94	7.25	0.58	3.45	0.02

Source: Self-elaboration.

Table 5.3 – Summary of the Olympus ensemble DSI-ESMDA results.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation				Avg.		Std. Dev.	
		Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.
5	No	0.89	0.28	0.96	0.26	2.90	2.70	21597697.52	1.42	26726853.22	0.01
	Yes	0.89	0.66	0.96	0.76	2.90	2.29	21597697.52	17.80	26726853.22	18.21
10	No	0.93	0.40	0.97	0.59	2.10	1.83	1584.40	0.98	1746.01	0.02
	Yes	0.93	0.59	0.97	0.85	2.10	1.56	1584.40	1.30	1746.01	0.29
15	No	0.96	0.41	0.97	0.26	2.71	2.05	325.51	1.10	273.67	0.02
	Yes	0.96	0.70	0.97	0.89	2.71	1.75	325.51	1.49	273.67	0.33
20	No	0.83	0.42	0.88	0.63	1.41	1.11	62.06	0.62	43.78	0.02
	Yes	0.83	0.57	0.88	0.86	1.41	1.09	62.06	0.72	43.78	0.06
25	No	0.96	0.36	0.97	0.27	1.81	1.64	9474346993.14	0.89	12035935454.92	0.02
	Yes	0.96	0.71	0.97	0.91	1.81	1.41	9474346993.14	5.49	12035935454.92	4.78
30	No	0.89	0.43	0.96	0.78	1.37	1.15	416.13	0.63	432.36	0.01
	Yes	0.89	0.60	0.96	0.87	1.37	1.10	416.13	0.75	432.36	0.09
35	No	0.89	0.42	0.92	0.45	1.73	1.49	293.01	0.81	266.32	0.01
	Yes	0.89	0.59	0.92	0.92	1.73	1.34	293.01	1.05	266.32	0.18
40	No	0.96	0.45	0.97	0.56	1.52	1.41	640.42	0.77	502.35	0.01
	Yes	0.96	0.72	0.97	0.88	1.52	1.24	640.42	1.76	502.35	0.74
45	No	0.92	0.39	0.93	0.34	2.37	2.07	567.02	1.10	524.09	0.01
	Yes	0.92	0.64	0.93	0.93	2.37	1.82	567.02	1.99	524.09	0.73
50	No	0.95	0.45	0.96	0.61	1.35	1.18	314.09	0.65	253.71	0.01
	Yes	0.95	0.70	0.96	0.91	1.35	1.12	314.09	1.05	253.71	0.23

Source: Self-elaboration.

Table 5.4 – Summary of the Olympus ensemble Gated DSI-ESMDA results ($\lambda = 2$).

Groundtruth realization	Loc.	Observation coverages History Pr.	Po.	Validation Pr.	Po.	MSE Pr.	Po.	Data mismatch Avg. Pr.	Po.	Std. Dev. Pr.	Po.
5	No	0.89	0.16	0.96	0.14	2.90	2.71	21597697.52	1.37	26726853.22	0.01
	Yes	0.89	0.66	0.96	0.76	2.90	2.26	21597697.52	17.80	26726853.22	18.21
10	No	0.93	0.29	0.97	0.47	2.10	1.84	1584.40	0.96	1746.01	0.01
	Yes	0.93	0.57	0.97	0.87	2.10	1.62	1584.40	1.31	1746.01	0.30
15	No	0.96	0.30	0.97	0.21	2.71	2.07	325.51	1.07	273.67	0.01
	Yes	0.96	0.67	0.97	0.89	2.71	1.75	325.51	1.48	273.67	0.34
20	No	0.83	0.32	0.88	0.55	1.41	1.14	62.06	0.61	43.78	0.01
	Yes	0.83	0.54	0.88	0.86	1.41	1.15	62.06	0.74	43.78	0.04
25	No	0.96	0.25	0.97	0.18	1.81	1.65	9474346993.14	0.84	12035935454.92	0.01
	Yes	0.96	0.69	0.97	0.91	1.81	1.39	9474346993.14	5.45	12035935454.92	4.76
30	No	0.89	0.34	0.96	0.71	1.37	1.17	416.13	0.63	432.36	0.01
	Yes	0.89	0.55	0.96	0.91	1.37	1.14	416.13	0.77	432.36	0.08
35	No	0.89	0.30	0.92	0.41	1.73	1.52	293.01	0.79	266.32	0.01
	Yes	0.89	0.53	0.92	0.92	1.73	1.40	293.01	1.06	266.32	0.18
40	No	0.96	0.29	0.97	0.34	1.52	1.41	640.42	0.73	502.35	0.01
	Yes	0.96	0.70	0.97	0.89	1.52	1.25	640.42	1.74	502.35	0.72
45	No	0.92	0.22	0.93	0.26	2.37	2.08	567.02	1.06	524.09	0.01
	Yes	0.92	0.59	0.93	0.93	2.37	1.83	567.02	1.97	524.09	0.72
50	No	0.95	0.33	0.96	0.47	1.35	1.19	314.09	0.63	253.71	0.01
	Yes	0.95	0.67	0.96	0.91	1.35	1.13	314.09	1.05	253.71	0.23

Source: Self-elaboration.

Table 5.5 – Summary of the real field ensemble DSI-ESMDA results.

Groundtruth realization	Loc.	Observation coverages History Pr.	Po.	Validation Pr.	Po.	MSE Pr.	Po.	Data mismatch Avg. Pr.	Po.	Std. Dev. Pr.	Po.
0	No	0.86	0.63	0.95	0.57	2.71	2.28	5.44	1.24	2.19	0.02
	Yes	0.86	0.63	0.95	0.57	2.71	2.28	5.44	1.24	2.19	0.02

Source: Self-elaboration.

Table 5.6 – Summary of the real field ensemble Gated DSI-ESMDA results ($\lambda = 2$).

Groundtruth realization	Loc.	Observation coverages History Pr.	Po.	Validation Pr.	Po.	MSE Pr.	Po.	Data mismatch Avg. Pr.	Po.	Std. Dev. Pr.	Po.
0	No	0.86	0.56	0.95	0.55	2.71	2.36	5.44	1.27	2.19	0.02
	Yes	0.86	0.57	0.95	0.58	2.71	2.36	5.44	1.27	2.19	0.02

Source: Self-elaboration.

6 Discussion

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Aims and structure

6.2 Summary of Key Findings

6.3 Interpretation of Findings

6.4 Evaluation of Existing Theories and Models

6.5 Limitations and Future Research

6.6 Chapter summary

Key takeaways

7 Conclusion

7.1 Summary of Findings

7.2 Contribution to Knowledge

7.3 Practical Implications

7.4 Limitations

7.5 Future Research

7.6 Closing Statement

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Appendices

Appendix A – Deriving the Innovation Gating Parameter

This chapter provides a self-contained derivation of the alpha parameter used in the innovation gating mechanism, as introduced in equation (3.12) of chapter 3. The derivation follows closely the work of Fortaleza *et al.* (2022), from where the gating mechanism was inspired.

Let $Y \in \mathbb{R}$ be a random variable representing the output of a given process in time, and Y_I the observation of Y. It is assumed that the observation is affected by a Gaussian noise $N \sim \mathcal{N}(0, \sigma_n^2)$, such that

$$Y_I = Y + N. \tag{A.1}$$

Let y_I be a realization of Y_I. Denoting by y_M the prediction of y from a given model, y_O is defined to be the adjusted prediction after the observation y_I is taken into account. For the sake of simplicity, y_O is given by the following exponential moving average (EMA) relationship:

$$\begin{cases} y_O[0] = y_I[0], & k = 0, \\ y_O[k] = \alpha[k] y_I[k] + (1 - \alpha[k]) y_M[k-1], & k > 0 \end{cases} \tag{A.2}$$

where $\alpha[k] \in [0, 1]$ is the smoothing factor of the EMA at time step index k.

Parameter $\alpha[k]$ represents the statistical confidence that the filter output reflects the same direction of the process output at sample k. More precisely, $\alpha[k]$ is given by the following expression:

$$\begin{aligned} \alpha[k] = \; & P(\text{sign}(Y_M[k-1] - Y_I[k]) = \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})) \\ & - P(\text{sign}(Y_M[k-1] - Y_I[k]) \neq \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})), \end{aligned} \tag{A.3}$$

where $Y_M[k-1]$ is an observation of Y_M with mean $\mu_{Y_M[k-1]}$ and Y_I an observation of Y_I with mean $\mu_{Y_I[k]}$. Here $\text{sign}(x)$ represents the sign function, which is defined as:

$$\text{sign}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases} \tag{A.4}$$

Equation (A.3) can be rewritten as follows:

$$\alpha[k] = 2P(\text{sign}(Y_M[k-1] - Y_I[k]) = \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})) - 1. \tag{A.5}$$

Defining a radom variable Y_{MI} whose observation at time k have mean and variance given by

$$\mu_{Y_{MI}[k]} = |\mu_{Y_M[k-1]} - \mu_{Y_I[k]}|, \tag{A.6}$$
$$\sigma^2_{Y_{MI}[k]} = \sigma^2_{Y_M[k-1]} + \sigma^2_{Y_I[k]} - 2\mathrm{Cov}(Y_M[k-1], Y_I[k]), \tag{A.7}$$

where $\mathrm{Cov}(Y_M[k-1], Y_I[k])$ is the covariance between $Y_M[k-1]$ and $Y_I[k]$, equation (A.5) can be expressed as follows:

$$\alpha[k] = 2P(Y_{MI}[k] > 0) - 1. \tag{A.8}$$

In order to find a practical implementation of $\alpha[k]$, equation (A.8) is analyzed under the following technical assumptions.

Assumption A.1. The random variable $Y_{MI}[k]$ can be approximated by a normal distribution with mean $\mu_{Y_{MI}[k]}$ and variance $\sigma^2_{Y_{MI}[k]}$.

Assumption A.2. Both high frequency and random behavior of the measurement noise implies that the observations $Y_M[k-1]$ and $Y_I[k]$ can be considered approximately statistically independent of each other. Mathematically, $\mathrm{Cov}(Y_M[k-1], Y_I[k]) \approx 0$.

Assumption A.3. The variance of the model's prediction is much smaller than the variance of the measured observation. Mathematically, $\sigma^2_{Y_M[k-1]} \ll \sigma^2_{Y_I[k]}$.

Assumption A.4. By considering an additive zero-mean measurement noise, the samples $y_M[k-1]$ and $y_I[k]$ can be used to estimate $\mu_{Y_M[k-1]}$ and $\mu_{Y_I[k]}$, respectively.

From assumption A.1, equation (A.8) can be rewritten as:

$$\begin{aligned} \alpha[k] &= 2P(\mathcal{N}(\mu_{Y_{MI}[k]}, \sigma^2_{Y_{MI}[k]}) > 0) - 1 \\ &= 2P(\mathcal{N}(0, \sigma^2_{Y_{MI}[k]}) \le \mu_{Y_{MI}[k]}) - 1, \end{aligned} \tag{A.9}$$

which can be expressed in terms of the Gauss error function:

$$\begin{aligned} \alpha[k] &= \mathrm{erf}\left(\frac{\mu_{Y_{MI}[k]}}{\sqrt{2\sigma^2_{Y_{MI}[k]}}}\right) \\ &= \mathrm{erf}\left(\frac{|\mu_{Y_M[k-1]} - \mu_{Y_I[k]}|}{\sqrt{2[\sigma^2_{Y_M[k-1]} + \sigma^2_{Y_I[k]} - 2\mathrm{Cov}(Y_M[k-1], Y_I[k])]}}\right). \end{aligned} \tag{A.10}$$

Applying assumption A.2, assumption A.3 and assumption A.4, equation (A.10) becomes

$$\alpha[k] = \operatorname{erf}\left(\frac{|y_M[k-1] - y_I[k]|}{\sqrt{2\sigma^2_{Y_I[k]}}}\right) = \operatorname{erf}\left(\frac{|y_M[k-1] - y_I[k]|}{\sqrt{2\sigma^2_n[k]}}\right). \tag{A.11}$$

Since knowledge of the measurement noise variance $\sigma^2_n[k]$ is not always available in practice, a fixed $\lambda \geq 1$ is introduced to scale the measurement noise standard deviation. Thus, the final expression for $\alpha[k]$ is given by

$$\alpha[k] = \operatorname{erf}\left(\frac{|y_M[k-1] - y_I[k]|}{\lambda\sqrt{2\sigma^2_n[k]}}\right) \tag{A.12}$$

which is equivalent to equation (3.12) in chapter 3.

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